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Study on Keystroke Dynamic with Feature of Pressure

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Abstract. This paper introduces a new database structure which is different from the existed one that only consists of press and release time. To improve, we add eigenvalue of pressure to it and we compare the results of new database structure with the old one to clarify whether the adding eigenvalue of pressure can improve the efficiency of the system. In addition, we come up a different keystroke dynamics method, Hidden Markov Models, to examine the efficiency of the new database structure and compare it with the results from another similar one, One-Class Support Vector Machine, to figure out whether Hidden Markov Models will behave better.

Introduction

The degree of informatization of modern society is rapidly deepening. Our life is fulfilled with all kinds of information. Therefore, the security problem should be paid more and more attention on. Nowadays pattern recognition technology has a plenty of applications and was applied in our daily lives. Among those methods, the most popular methods are Bio-metric authentication methods such as face recognition, fingerprint identification and palm print recognition [4]. However, all of them need complicated external devices. On the contrast, keystroke dynamics authentication is a simple way to authenticate without additional hardware [5,14].

Research on keystroke dynamics was started long time before, but it is still an on-going research. Traditional database of keystroke dynamics only contains press time and release time, which display the rhythm of the typing manner. However, many smart phones already have the ability to test the pressure of people when they are typing, so we add the new feature of the keystroke pressure on the basis of old features and hope to observe the improvement of the recognition rate. Hidden Markov Models are applied as a method of keystroke dynamics to compare the efficiency with One-Class Support Vector Machine to find out the better way to identify the characters on keyboard.

Keystroke Dynamic

Data Preparation

Our experiment data consists of two parts: time and pressure. It is presented in the form of a sequence: $P_1,R_1,P_2,R_2,\ldots,P_m,R_m,T_1,T_2,\ldots,T_m$.

P_i (i=1,2,…,m) represents the press time, which is defined as the time when user stroke the ith key. R_i represents the release time, which is defined as the time when user release the ith key [1]. T_i represents the pressure value when the ith key is pressed. From this set of raw data series, we can extract three types of features: flight time, dwelling time and the key pressure. So the time of keystroke duration is R_i-P_i (dwelling time), the time of keystroke interval is P_{i+1}-R_i (flight time) [6] and the key pressure is T_i. As a result, the sequence extracted from the raw data is:

$$F(n) = (R_1\text{-}P_1, P_2\text{-}R_1, R_2\text{-}P_2, ..., P_m\text{-}R_{m-1}, R_m\text{-}P_m, T_1, T_2, ..., T_m). \quad (1)$$

Our Experiments are performed on the extracted feature sequences and are examined in two methods, Hidden Markov Models and One-Class Support Vector Machine.

Recognition Methods

SVM (*Support Vector Machine*)

Support vector machine (SVM) is widely used in the field of machine learning, especially the field of unsupervised learning [8]. In our paper, we use the One-Class SVM from LIBSVM library [3] as a tool to test the efficiency of it on our data. One-class SVM has advantages in dealing with the small sample size, high dimension data. In training, the SVM separates the training samples by finding a hyper plane to maximizes the distance between the positive examples and the negative ones to achieve the purpose of classification [2,13].

HMM (*Hidden Markov Model*)

Hidden Markov Models (HMM) is based on the Markov chain which is commonly used in speech recognition [9,10,11]. HMM based system is also widely used in pattern recognition, and it performed well in some fields. Hidden Markov model is suitable for the process whose transition cannot be observed. We use VITERBI algorithm to make the best estimate of the state sequence by observing the outputs of sequence. We can get different states sequences from different users and to accomplish the purpose of identifying different users.

Experiments and Results

In the experimental part, we use OC-SVM and HMM to identify collected data and the results are analyzed as following.

Results of SVM

In the first part, we adjust the threshold in the SVM algorithm to record the True Positive Rate (TPR) and False Positive Rate (FPR) under different thresholds. The True Positive Rate is the rate that the true original user-entered password is accepted as correct one and the False Positive Rate is the rate that other users are accepted by mimicking the patterns of entering the same password [12]. We plot the ROC curve of the two rates whose abscissa is FPR and the ordinate is TPR. In order to compare the varieties of system recognition rate of three different kinds of sequences, we plot three ROC curves at the same coordinates. Three curves are named separately: series 1 (The feature sequence contains both time series and pressure eigenvalues), series 2 (The feature sequence only contains the time eigenvalues), and series 3 (the feature sequence only contains the pressure eigenvalues).

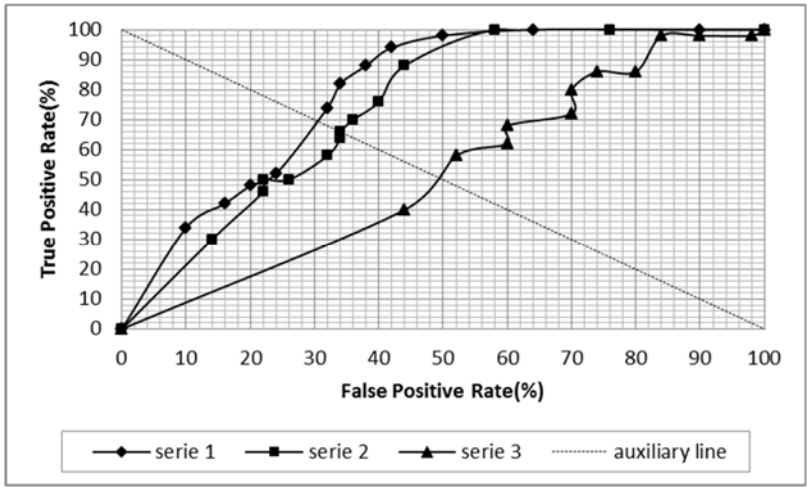

Figure 1. Comparison of the ROC results of three kinds of database based on SVM.

From Fig.1, we can get the value of Equal Error Rate (EER) [6]. It can be concluded that the effect of sequence recognition with time and pressure features is better than that only applying time feature or pressure feature value, and the addition of a pressure signature of the keystroke dynamics does improve the recognition rates. Table 1 shows the recognition rates for different thresholds, where *n* is the threshold.

Table 1. TPR and FPR of three kinds of database with different thresholds.

Feature	Time and pressure		Time only		Pressure only	
n	TPR	FPR	TPR	FPR	TPR	FPR
0.1	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
0.3	100.0%	100.0%	100.0%	76.0%	100.0%	100.0%
0.35	100.0%	90.0%	100.0%	58.0%	98.0%	98.0%
0.45	100.0%	64.0%	88.0%	44.0%	98.0%	90.0%
0.5	98.0%	50.0%	76.0%	40.0%	98.0%	84.0%
0.55	94.0%	42.0%	70.0%	36.0%	86.0%	80.0%
0.6	88.0%	38.0%	66.0%	34.0%	86.0%	74.0%
0.65	82.0%	34.0%	64.0%	34.0%	80.0%	70.0%
0.7	74.0%	32.0%	58.0%	32.0%	72.0%	70.0%
0.8	52.0%	24.0%	50.0%	26.0%	68.0%	60.0%
0.85	48.0%	20.0%	50.0%	22.0%	62.0%	60.0%
0.9	42.0%	16.0%	46.0%	22.0%	58.0%	52.0%
0.99	34.0%	10.0%	30.0%	14.0%	40.0%	44.0%
1	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Results of HMM

In the second part, we evaluate the performance of HMM. Notice that the training model for each state is composed of the linear combinations of several Gaussian probability density function (PDF), different number and weight of Gaussian PDF leads to different recognition rate. Like what we did in the first part of experiments, three different kinds of feature sequences, combination of time and pressure, only time and only pressure are tested. Table 2 shows the different recognition rate under different Gaussian PDF. At the same time, the recognition rate fluctuates a little because of the K-means clustering algorithm. The results are averaged over 50 independent experiments with the same weight series and the variance is given.

In Table 2, "Series of weight" represents the coefficients of Gaussian PDF. For example, the first series of first column is consisted of four digits 3222, which means that the model is composed of four PDFs, in which the weight of the first PDF is 3/(3+2+2+2), the weight of the second PDF is 2/(3+2+2+2), and the rest can be done as the same manner.

Table 2. Recognition rate and variance based on HMM with different PDFs.

HMM								
Combination of time and pressure			Only time			Only pressure		
Series of weight	Recognition rate	variance	Series of weight	Recognition rate	variance	Series of weight	Recognition rate	variance
3222	0.6452	0.007103	2222	0.5926	7.41E-05	3 3	0.3454	0.001446
3221	0.6362	0.005012	3222	0.5784	0.000172	2 2	0.4	0.003343
2221	0.6136	0.00553	3322	0.5662	0.000257	2 1	0.3676	0.001292
2222	0.5938	0.009412	2221	0.5556	5.00E-05	1 1	0.29	1.54E-31
3321	0.5608	0.007746	2211	0.553	8.09E-05	1 2	0.348	0.001457
2111	0.4444	0.002188	2111	0.5354	0.000144			

According to the results in Table 2, the recognition performance of feature sequence with both of time and pressure information is better than feature sequences only consist of time or pressure information. It means that in HMM algorithm, we can also improve the accuracy of keystroke dynamics authentication by taking characteristic of pressure into consideration.

Comparison

In the third part, we compare the highest recognition rate of HMM and SVM (from table 2 and table 3), the recognition rates of SVM are showed in Table 3. We can see that the SVM algorithm is a little better than HMM algorithm.

Table 3. Recognition rate of three kinds of database based on SVM with different thresholds.

SVM	Time+Pressure	Time	Pressure
n	Recognition Rate	Recognition Rate	Recognition Rate
0.10	50.0%	50.0%	50.0%
0.30	50.0%	62.0%	50.0%
0.35	55.0%	71.0%	50.0%
0.45	68.0%	72.0%	54.0%
0.50	69.0%	68.0%	57.0%
0.55	76.0%	67.0%	53.0%
0.60	75.0%	66.0%	56.0%
0.65	74.0%	65.0%	55.0%
0.70	71.0%	63.0%	51.0%
0.80	64.0%	62.0%	54.0%
0.85	64.0%	64.0%	51.0%
0.90	63.0%	62.0%	53.0%
0.99	62.0%	58.0%	48.0%
1.00	50.0%	50.0%	50.0%

Conclusion and Future Work

By analyzing the results of two algorithms, we can find that the recognition rate are both improved after adding the feature of pressure. Therefore, it is a feasible method to improve the recognition efficiency by adding the pressure feature in the process of keystroke dynamics authentication.

Our future work will focus on collecting a larger database to do the experiments. Also, we plan to test other recognition methods based on the sequence which consists of feature of time and pressure as well.

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